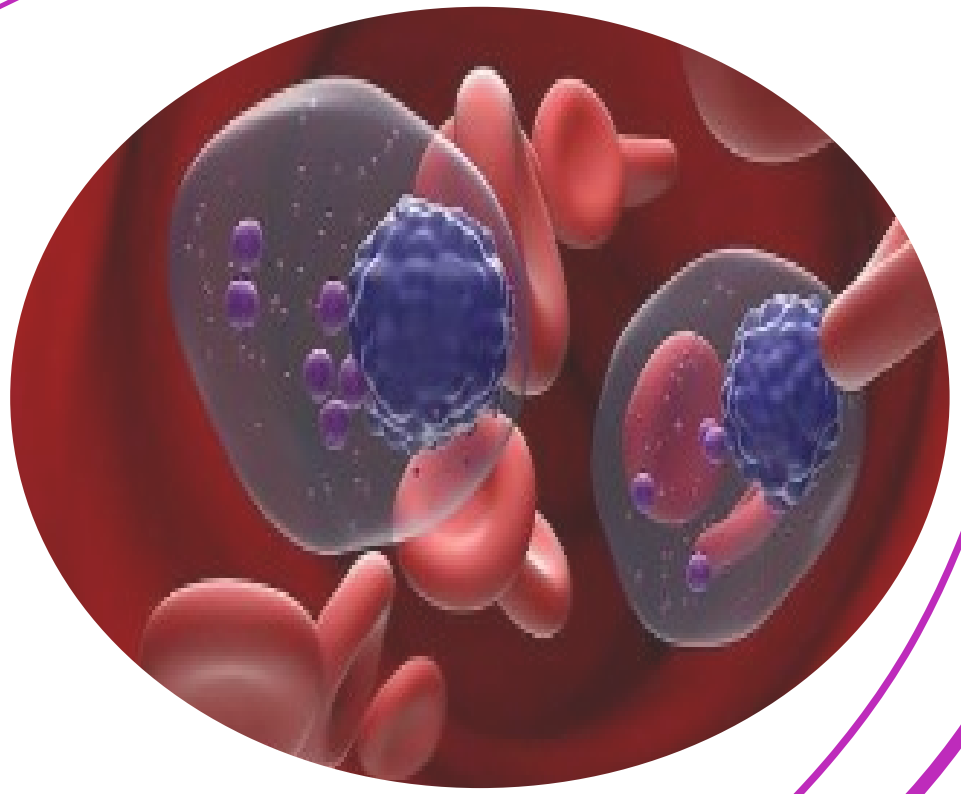


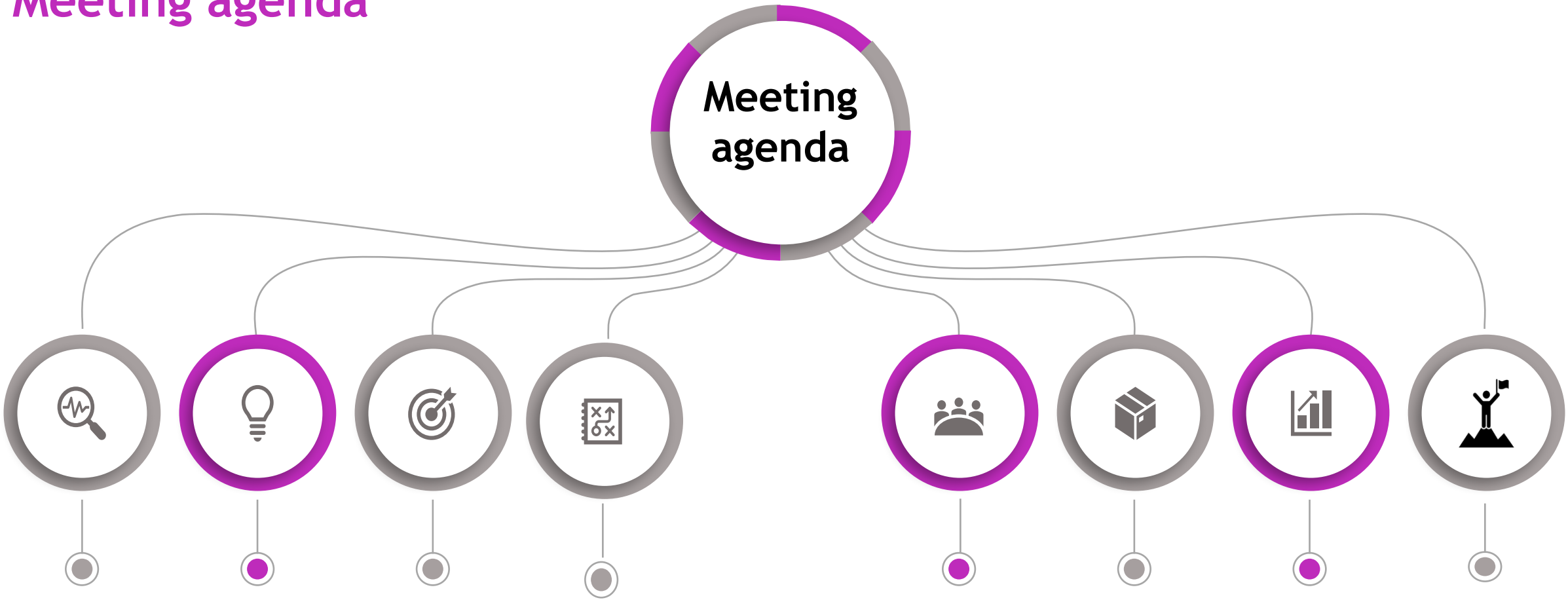
BMS CELMoD Advisory Board Meeting

UAE

Virtual Advisory Board Meeting

27 June 2026

Meeting agenda



Advisors

Advisors	Affiliation	Country
Asma Olama	Dubai Hospital, Dubai	UAE
Arif Alam	Tawam Hospital, Al Ain	UAE
Hani Yousif Osman	Tawam Hospital, Al Ain	UAE
Faraz Khan	American Hospital, Dubai	UAE
Kourosh Lotfi	Sheikh Shakhbout Medical City, Abu Dhabi	UAE
Hasan Al Yaseen	Dubai Hospital, Dubai	UAE
Emmanouil Nikolousis	Yas Clinic, Abu Dhabi	UAE
Panayotis Kaloyannidis	Burjeel Cancer Institute, Abu Dhabi	UAE

BMS Attendees



Mina Nagib
Senior Scientific Advisor, MEA



Ahmed Belal
Senior Product Manager, Gulf countries



Mayada Al Ahmad
General Manager, GCC



Mohamed El-Halfawi
Oncology & Hematology Franchise lead, Gulf Countries



Meeting Introduction: Moderator

Prof. Dr. Mahmoud Marashi

Head, Academic Department of Medicine at Dubai Medical College/Dubai Health Authority.

Dr Marashi is a Consultant Hematologist at Mediclinic City Hospital, Dubai, with over 30 years of experience in clinical hematology.

He is recognized as one of the pioneers of hematology services in the UAE, having established the clinical hematology service within Dubai Health Authority. His clinical expertise covers both benign and malignant blood disorders, including leukaemia, myeloma, and thrombosis-related conditions.

He has also been actively involved in medical education for more than 20 years. He is a Professor of Medicine and Haematology and serves as the Head of the Academic Department of Medicine at Dubai Medical College/Dubai Health Authority.



Meeting Introduction: Speaker

Prof. Paul Richardson

RJ Corman Professor of Medicine at Harvard Medical School, USA.

He is currently the RJ Corman Professor of Medicine at Harvard Medical School, as well as Clinical Programme Leader and Director of Clinical Research at the Dana-Farber Cancer Institute (DFCI).

His leadership in landmark Phase III trials has directly shaped global standards of care. These include the VISTA trial, DETERMINATION (IFM/DFCI) study, OPTIMISMM trial, ICARIA trial, and GRIFFIN study.

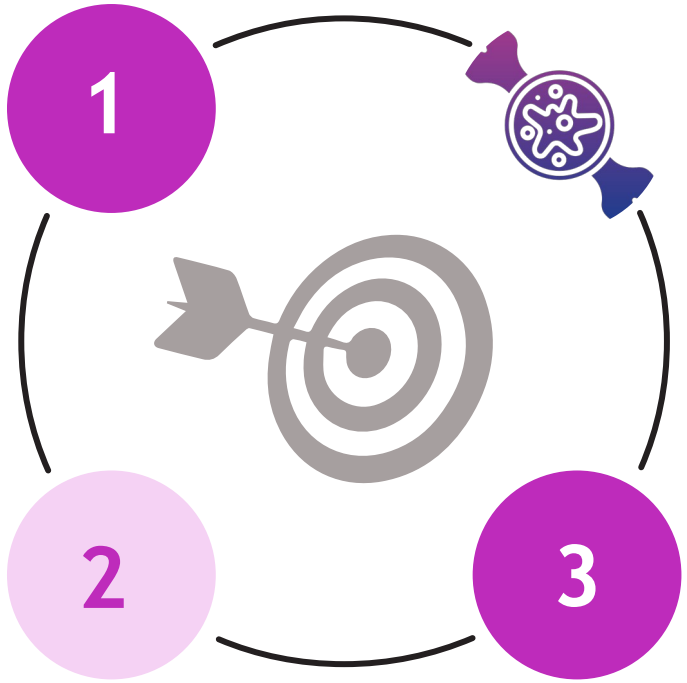
He has led the clinical development of many of today's foundational myeloma therapies, including thalidomide, lenalidomide, bortezomib, pomalidomide, daratumumab, elotuzumab, isatuximab and panobinostat, as well as the more recent oral cereblon E3 ligase modulator (CelMoD), mezigdomide.

Dr. Richardson has authored or co-authored 490+ original articles and 430+ reviews and editorials in peer-reviewed journals and is consistently ranked among the top 1% of most-cited scientists globally.

Advisory Board Objectives

1. Assess the burden of multiple myeloma and identify key unmet medical needs in the region

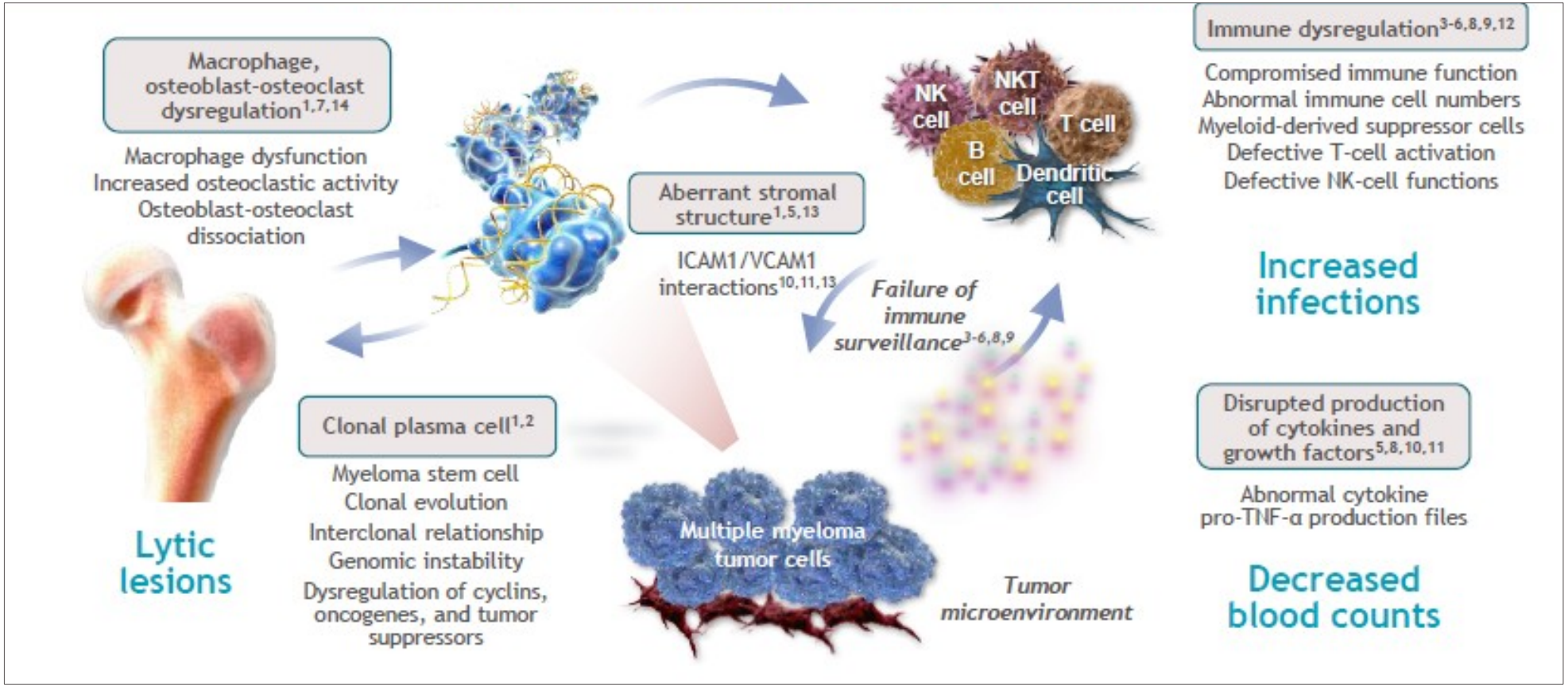
2. Gather expert insights on the CELMoDs mechanism of action and clinical development program



3. Explore the potential role and clinical value of Iberdomide and Mezigdomide within the evolving multiple myeloma treatment landscape

Session 1: MM disease burden/unmet medical needs and current treatment landscape

MM is characterized is not driven solely by genetic abnormalities within the myeloma cell, but also by the surrounding pro-tumor microenvironment.



Understanding inflammasome biology is important because inflammation actively drives disease progression, immune dysfunction and treatment response. The Duffy-null phenotype may be particularly relevant in Middle Eastern populations and could be considered during patient assessment where clinically appropriate.

ICAM1: intercellular adhesion molecule 1; **MM:** multiple myeloma; **NK:** natural killer; **NKT:** natural killer T; **TNF-α:** tumor necrosis factor alpha; **VCAM1:** vascular cell adhesion molecule 1.

1. Raab MS, et al. Lancet 2009;374:324–339; 2. Morgan GJ, et al. Nat Rev Cancer 2012;12:335–348; 3. Katodritou E, et al. Am J Hematol 2011;86:967–973; 4. Braga WMT, et al. Clin Dev Immunol 2012;2012:293479; 5. Pratt G, et al. Br J Haematol 2007;138:563–579; 6. Rosenblatt J, et al. J Immunother 2011;34:409–418; 7. Kyle RA, Rajkumar SV. N Engl J Med 2004;351:1860–1873; 8. Cook G, Campbell JD. Blood Rev 1999;13:151–162; 9. Bernal M, et al. Hum Immunol 2009;70:854–857; 10. Gupta D, et al. Leukemia 2001;15:1950–1961; 11. Jourdan M, et al. Eur Cytokine Netw 1999;10:65–70; 12. Favaloro J, et al. Leuk Lymphoma 2014;55:2893–2900; 13. Damiano JS, et al. Blood 1999;93:1658–1667; 14. Oranger A, et al. Clin Dev Immunol 2013;2013:289458.

Current treatment landscape

Quadruplet regimens remain a key frontline approach, while ASCT continues to be standard for transplant-eligible patients.

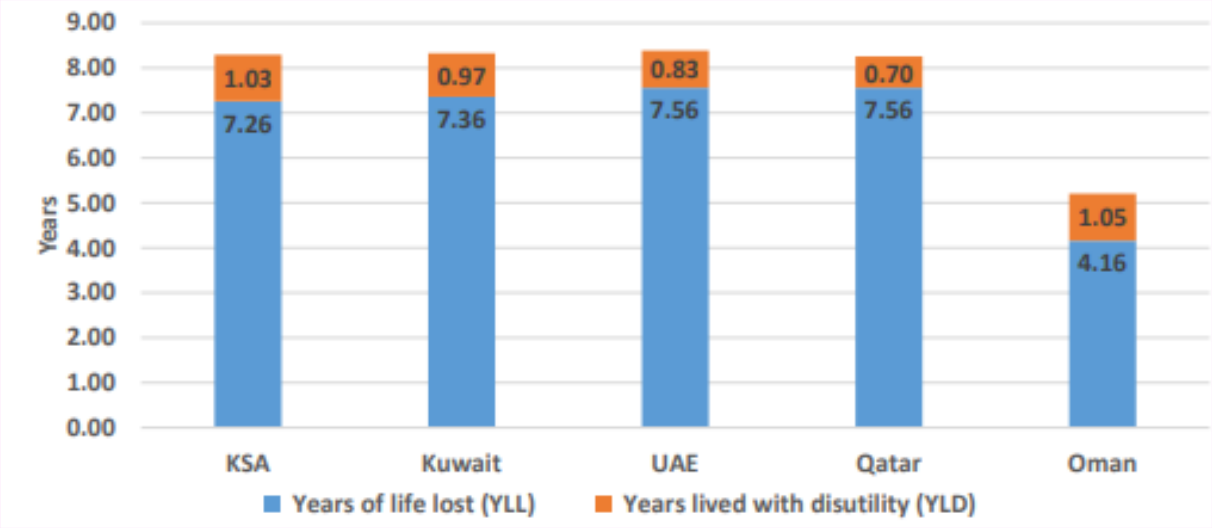
MRD-driven strategies may support more selective or deferred use of transplant in patients achieving deep responses.

ADCs, CAR T-cell therapies, bispecific antibodies and GPRC5D-targeted therapies are expanding later-line options.

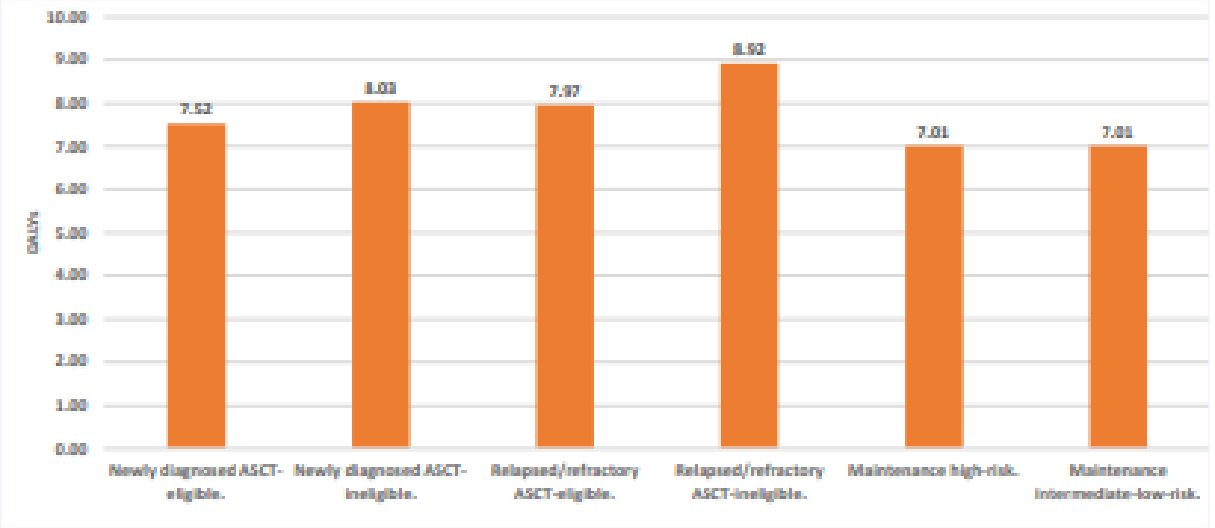
ADCs: antibody-drug conjugates; **ASCT:** autologous stem cell transplantation; **CAR T:** chimeric antigen receptor T-cell; **GPRC5D:** G protein-coupled receptor class C group 5 member D; **MRD:** minimal residual disease.

MM remains incurable in ~40% of patients (potentially up to ~60% for long-term disease control)

DALYs per patient with MM[1]



Average DALYs per patient per classification[1]

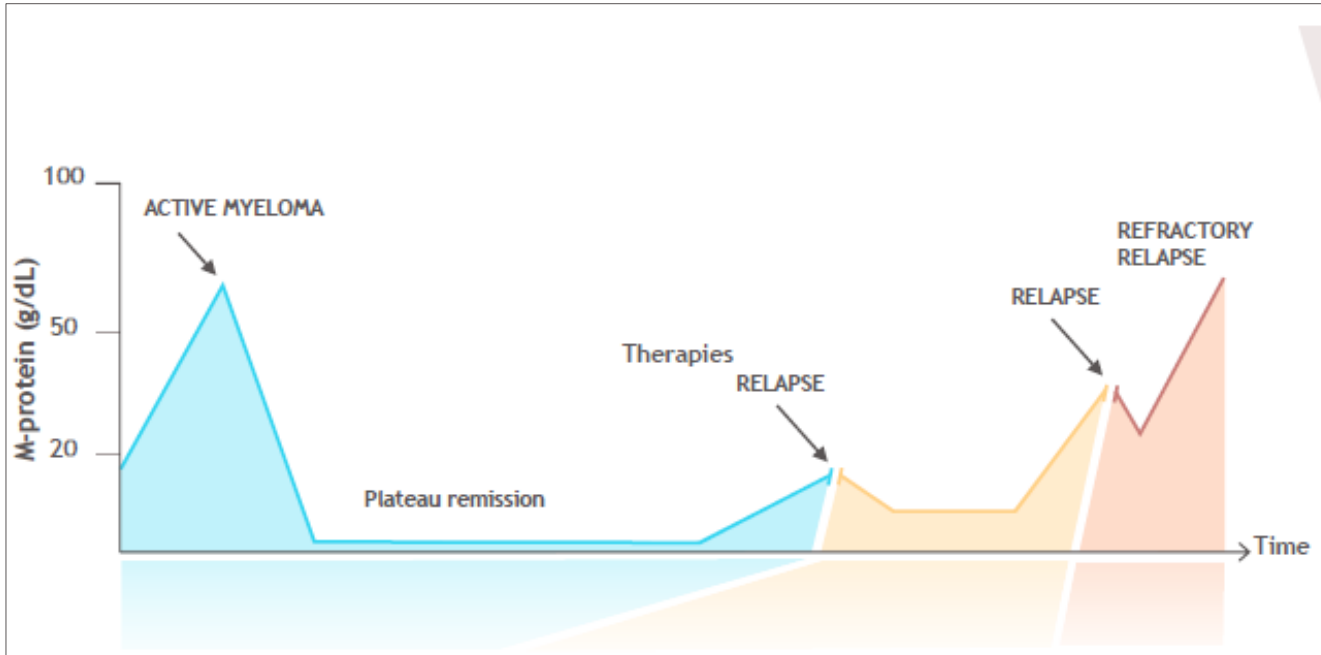


MM imposes a significant burden on the Gulf region, especially the economic burden. The relapsed/refractory ASCT-ineligible group recorded the highest DALYs across all groups while the relapsed/refractory ASCT-eligible group incurred the highest direct medical costs[1]

ASCT: autologous stem cell transplantation; **DALY:** disability-adjusted life year; **KSA:** Kingdom of Saudi Arabia; **MM:** multiple myeloma; **UAE:** United Arab Emirates; **YLD:** years lived with disability; **YLL:** years of life lost.

1. Farghaly M, et al. ISPOR Europe. Poster EE178.

Unmet needs persist despite advances with IMiDs, PIs, anti-CD38 antibodies and BCMA-targeted therapies





Significant unmet needs remain

- More functional cure and deep sustained responses
- Improved tolerability and QoL
- Fixed durations or treatment-free intervals
- Enhanced outcomes for high unmet need patient populations

BCMA: B-cell maturation antigen; **EMD:** extramedullary disease; **IMiDs:** immunomodulatory drugs; **MM:** multiple myeloma; **PIs:** proteasome inhibitors; **QoL:** quality of life; **RRMM:** relapsed/refractory multiple myeloma.

1. Hajek R. Strategies for the treatment of multiple myeloma in 2013: Moving toward the cure. InTech eBooks. Published online April 10, 2013. https://doi.org/10.5772/55366.

Real-world patients with lenalidomide-refractory MM and 1 to 3 prior LOT progress rapidly through therapies

- Outcomes remain suboptimal in lenalidomide-refractory MM, with median PFS of approximately 6.3 to 6.5 months regardless of prior treatment lines.
- Lenalidomide-exposed but non-refractory patients had a longer, although still modest, median PFS of 9.2 months.

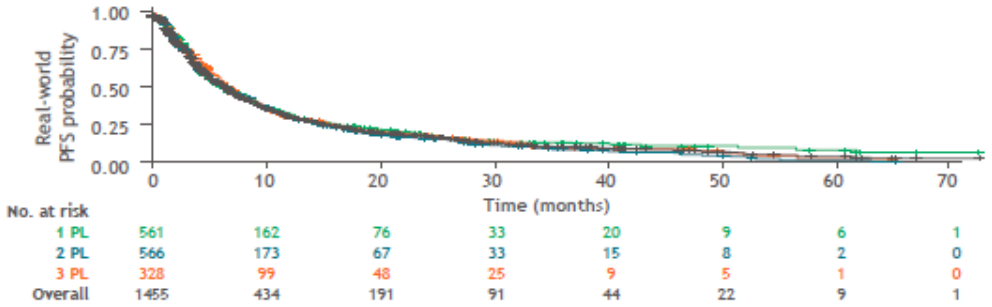
Flatiron US, N=1455[1]

Patients treated between Jan 2016 and April 2022
Note: No patients were treated with CAR T

Immediate next treatment regimens received after patients became LEN-refractory (NCCN-preferred regimens)

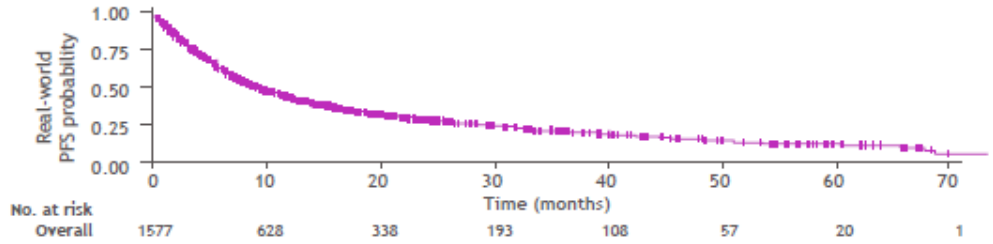
Treatment regimen, N (%)	Overall (N=1455)
DPd	192 (13.2)
KPd	104 (7.1)
DVd	95 (6.5)
DKd	56 (3.8)
PVd	39 (2.7)
IxaPd	14 (1.0)
IsaKd	3 (0.2)
IsaPd	3 (0.2)
Total	506 (34.8)

Median PFS in LEN-refractory patients (N = 1455)



Median (95% CI)
Overall 6.5 (5.9-7.0)
1 prior LOT 6.4 (5.2-7.4)
2 prior LOT 6.3 (5.5-7.3)
3 prior LOT 6.5 (5.8-7.8)

Median PFS in LEN-exposed-not-refractory patients (N = 1577)

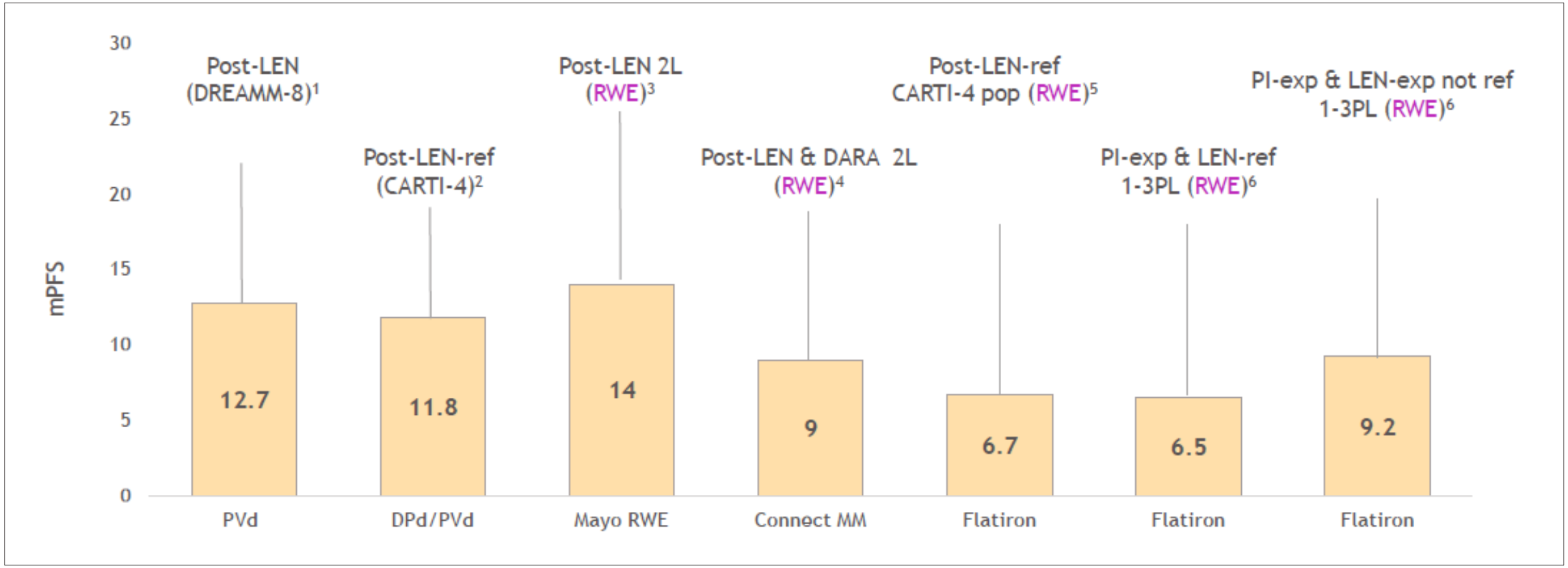


Median (95% CI)
Overall 9.2 (8.5-10.6)

CAR T: chimeric antigen receptor T-cell therapy; **CI:** confidence interval; **DKd:** daratumumab, carfilzomib, and dexamethasone; **DPd:** daratumumab, pomalidomide, and dexamethasone; **DVd:** daratumumab, bortezomib, and dexamethasone; **IsaKd:** isatuximab, carfilzomib, and dexamethasone; **IsaPd:** isatuximab, pomalidomide, and dexamethasone; **IxaPd:** ixazomib, pomalidomide, and dexamethasone; **KPd:** carfilzomib, pomalidomide, and dexamethasone; **LEN:** lenalidomide; **LOT:** line of therapy; **MM:** multiple myeloma; **N:** number of patients; **NCCN:** National Comprehensive Cancer Network; **PFS:** progression-free survival; **PL:** prior line of therapy; **PVd:** pomalidomide, bortezomib, and dexamethasone; **RWE:** real-world evidence.

1. Dhakal, B, et al. Blood Adv. 2024;8(19):5062

Median PFS remains modest at 6.5 to 14 Months in lenalidomide-exposed or lenalidomide-refractory populations



2L: second-line therapy; **DARA:** daratumumab; **DPd:** daratumumab, pomalidomide, and dexamethasone; **exp:** exposed; **LEN:** lenalidomide; **MM:** multiple myeloma; **mPFS:** median progression-free survival; **PI:** proteasome inhibitor; **PL:** prior line of therapy; **pop:** population; **PVd:** pomalidomide, bortezomib, and dexamethasone; **ref:** refractory; **RWE:** real-world evidence.

1. Dimopoulos, M, et al. NEJM2024; 2. San Miguel, J, et al. NEJM2023; 3. Goel, U. et al. Blood Cancer J. 2024;14(1):55; 4. Cowan, AJ, et al. IMS 2023; 5. Touzeau C, et al. EHA 2024; Poster P967; 6. Dhakal, B, et al. Blood Adv. 2024;8(19):5062.

KarMMa-3 is the only prospective, randomized phase 3 study in which nearly all patients had prior lenalidomide exposure, and all had prior anti-CD38 exposure

Prior therapy exposure across relapsed or refractory MM trials

		KPd SELECT[1]	SVd BOSTON[2]	IxaRd TOURMALINE-MM1[3]	DKd CANDOR[4,5]	IsaKd IKEMA[6,7]	ERd ELOQUENT-2[8]	KRd ASPIRE[9]	Cilta-cel CARTITUDE-4[10]	Kd ENDEAVOR[11]	PVd OPTIMISMM[12]	DRd POLLUX[13,14]	DVd CASTOR[15]	DPd APOLLO[16]	BVd DREAMM-7[17]	BPd DREAMM-8[18]	IsaPd ICARIA-MM[19]	EPd ELOQUENT-3[20]	Ide-cel KarMMa-3[21,22]	Elra MagnetisMM-3[23]	Talq MonumenTAL-1[24]	Tec MajesTEC-1[25]	Linvo LINKER-MM1[26]
Trial Phase		Ph2, single arm	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph3	Ph2, random	Ph3	Ph2, single arm	Ph1-2	Ph1-2	Ph1-2
Number of prior LOT requirement		1-2	1-3	1-3	1-3	1-3	1-3	1-3	1-3	1-3	1-3	≥1	≥1	≥1	≥1	≥1	≥2	≥2	2-4	≥3	≥3	≥3	≥3
Prior therapy requirements	IMiDs	LEN	-	-	-	-	-	-	IMiD	-	LEN	-	-	LEN	-	LEN	LEN	LEN	IMiD	IMiD	IMiD	IMiD	IMiD
	Anti-CD38	Anti-CD38*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Anti-CD38	Anti-CD38	Anti-CD38	Anti-CD38	Anti-CD38
Patients with prior exposure, %	LEN	100	39	12[27]	39	40	5[28]	20	100	38	100	18	36	100†	52	100	100	98	98	100†	100†	100†	100†
	Anti-CD38	77	6	0	<1	3	0	-	25	-	-	-	-	-	1	25	-	2	100	100	100	100	100

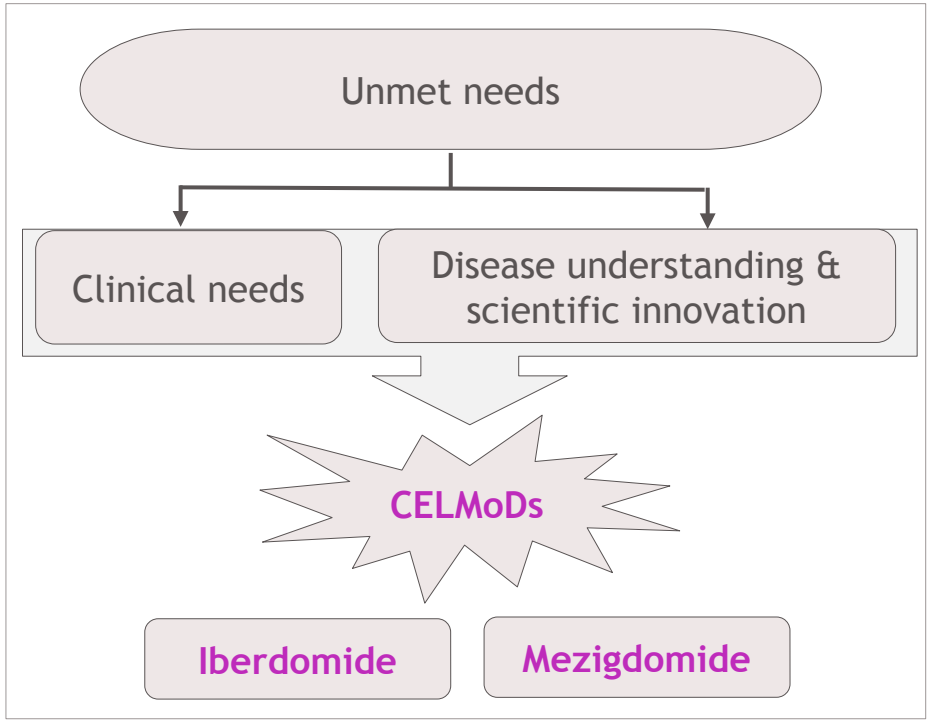
No treatments were studied in 2L patients with required prior LEN and an anti-CD38 antibody exposure

*At study initiation, included patients must have completed at least two consecutive cycles of daratumumab; this criterion was removed in March 2021, approximately 8 months before the last patient enrolled. †IMiD exposure.
See speaker notes for abbreviations and reference.

Need for CELMoDs

- Immune exhaustion and treatment failure across IMiDs, PIs, anti-CD38 antibodies, and BCMA-targeted therapies highlight persistent unmet needs.
- Lenalidomide- and anti-CD38-exposed or refractory populations remain a key treatment challenge.
- Outcomes are particularly poor in older, frailer patients and those exposed or refractory to both lenalidomide and anti-CD38 therapy, with anti-CD38 retreatment yielding a PFS of only 4 to 5 months.
- Differentiated therapies are needed to overcome resistance and restore immune function.
- CELMoDs (iberdomide and mezigdomide) offer direct anti-myeloma activity together with immune activation.
- Durable, safe and convenient oral therapies are needed, particularly for older patients and those with EMD or triple-class refractory MM.

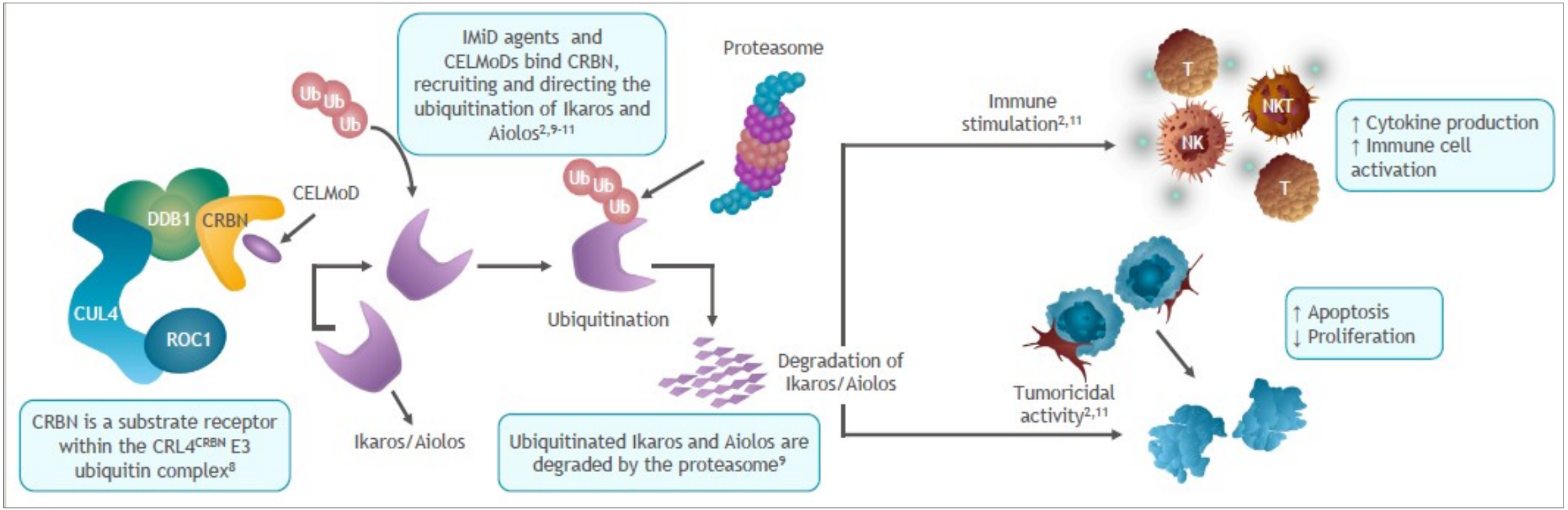
Integration of CELMoDS in MM treatment landscape



anti-CD38: anti-cluster of differentiation 38 monoclonal antibodies; **BCMA:** B-cell maturation antigen; **CELMoDs:** cereblon E3 ligase modulators; **EMD:** extramedullary disease; **IMiDs:** immunomodulatory drugs; **MM:** multiple myeloma; **PFS:** progression-free survival; **PIs:** proteasome inhibitors.

CELMoDs: Mechanism of action

CELMoDs (iberdomide, mezigdomide) are oral CRBN E3 ligase modulators that degrade Ikaros/Aiolos, resulting in antitumor and immunostimulatory effects; mezigdomide shows the highest CRBN engagement (~100%) and the most potent immune activation.[1-7]



CRBN: cereblon; **CELMoDs:** cereblon E3 ligase modulators; **CRL4CRBN:** CUL4-ROC1-DDB1-CRBN; **CUL4:** cullin-4; **DDB1:** DNA damage-binding protein 1; **IMiD:** immunomodulatory drug; **NK:** natural killer; **NKT:** natural killer T; **ROC1:** regulator of cullins 1; **T:** T-cell; **Ub:** ubiquitin.

1. Matyskiela ME et al. J Med Chem. 2018;61(2):535-542; 2. Bjorklund CC et al. Leukemia. 2020;34(4):1197-1201; 3. Carrancio S et al. Blood. 2021;138(suppl 1):1200; 4.Lopez-Girona A et al. Poster presentation at ICML 2021 [abstract 232]; 5. Carrancio S et al. Poster presentation at ASH 2021 [abstract 1200]; 6. Richardson PG et al. Blood. 2021;138(suppl 1):2731; 7. Wong L, et al. Oral presentation at ASCO 2020 [abstract 8531]; 8. Fischer ES et al. Nature. 2014;512:49-53; 9. Krönke J et al. Science. 2014;343:301-305; 10. Gandhi AK, et al. Br J Haematol. 2014;164(6):811-821; 11. Hansen JD et al. J Med Chem. 2020;63(13):6648-6676.

Discussion 1:
Epidemiology and local practice

What is the number of new multiple myeloma cases you see in your center annually?

Regional clinical experience shared by experts



UAE
~10-20 new MM cases annually at each centre

- Case volumes vary by practice setting, with around 6-8 new patients per year in a single private practice
- Estimated national incidence was approximately 50-60 new MM cases per year

MM: multiple myeloma; **UAE**: United Arab Emirates.

What is your current treatment approach for patients with multiple myeloma?

Regional clinical experience shared by experts

- Daratumumab-based quadruplet therapy, particularly DVRd, is the preferred frontline approach for most fit patients, followed by consolidation and daratumumab plus lenalidomide maintenance.
- In transplant-ineligible patients, treatment is tailored according to frailty: fitter patients may receive quadruplet therapy, whereas frailer patients are generally treated with reduced-intensity triplets such as DRd or DVd.
- Autologous transplant remains standard for eligible patients, although MRD status is increasingly influencing transplant timing and decision-making.
- Bortezomib remains the preferred frontline PI, while carfilzomib is generally reserved for later lines or selected younger, high-risk patients because of cardiovascular toxicity concerns.
- Treatment sequencing is increasingly guided by risk profile, tolerability, MRD response and access to therapy, with emerging combinations expected to further refine the role of PIs.



DRd: daratumumab, lenalidomide, and dexamethasone; **DVRd:** daratumumab, bortezomib, lenalidomide, and dexamethasone; **DVd:** daratumumab, bortezomib, and dexamethasone; **MRD:** minimal residual disease; **PIs:** proteasome inhibitors.

What are the preferred treatment options for patients with multiple myeloma in the second- and third-line settings?

Preferred second- and third-line treatment approaches

Treatment selection is guided by duration of prior response, previous drug exposure, patient fitness, and access to advanced therapies

- **In the second-line setting,** carfilzomib-based combinations are commonly used after frontline DVRd, while rechallenge with prior therapy may be considered following a prolonged remission.
- Belantamab-based ADC combinations are increasingly used in relapsed disease because of their activity, manageable safety profile, and potential for combination with mezigdomide.
- **In the third-line setting and beyond,** options include BPd, BVd, bispecific antibodies, and CAR T-cell therapy, although careful patient selection is required because of infection, neurological, and treatment-related risks.
- Selinexor remains an underused option and may be effective at low weekly doses, particularly in combination with mezigdomide, with proactive management of cytopenias using G-CSF or romiplostim.
- Mezigdomide was viewed as a flexible backbone across multiple treatment lines, with dose optimization and supportive care considered essential.

ADC: antibody-drug conjugate; **BPd:** bortezomib, pomalidomide, and dexamethasone; **BVd:** bortezomib, and dexamethasone; **CAR T-cell:** chimeric antigen receptor T-cell therapy; **DVRd:** daratumumab, bortezomib, lenalidomide, and dexamethasone; **G-CSF:** granulocyte colony-stimulating factor.

Guideline use in clinical practice

- ✓ NCCN and regional Gulf guidelines are commonly used to guide treatment decisions.
- ✓ Some centres also refer to mSMART recommendations.
- ✓ Guideline implementation is influenced by drug availability, insurance coverage, and local practice constraints.
- ✓ Advisors highlighted the importance of maintaining and publishing updated region-specific guidelines that reflect real-world clinical practice.

mSMART: Mayo stratification for myeloma and risk-adapted therapy; **NCCN**: National Comprehensive Cancer Network.

Role of high-dose melphalan

- ✓ High-dose melphalan should be used selectively, particularly in very young patients, because of its potential genetic toxicity and risk of secondary malignancies.
- ✓ In standard-risk, MRD-negative younger patients, stem-cell collection with deferred transplant may be considered instead of immediate transplantation.
- ✓ Long-term lenalidomide may also increase MDS risk; however, available evidence suggests a higher incidence of secondary malignancies with high-dose melphalan than with lenalidomide alone.

MDS: myelodysplastic syndrome; **MRD:** minimal residual disease.

Key takeaways

- MM case volume varies across centres, with some advisors reporting around 10–20 new cases annually.
- DVRd was commonly identified as the preferred first-line approach for most fit patients followed by consolidation and daratumumab-lenalidomide maintenance.
- In transplant-ineligible patients, treatment is tailored by frailty, with triplets for frailer patients and quadruplets for fitter patients.
- The role of transplant is evolving. While BMT remains standard practice in some centres, MRD assessment is increasingly influencing decisions.
- PIs remain an important backbone with bortezomib continues to be widely used, with weekly dosing, supportive care and dose escalation helping to reduce neuropathy.
- Carfilzomib was favored in younger or ultra-high-risk patients, but its use remains cautious due to cardiovascular toxicity and is often reserved for later lines.

BMT: bone marrow transplantation; **DVRd:** daratumumab, bortezomib, lenalidomide, and dexamethasone; **MM:** multiple myeloma; **MRD:** minimal residual disease; **PIs:** proteasome inhibitors.

Key takeaways (Contd.)

- In relapsed/refractory settings, treatment decisions remain highly individualized and depend on prior therapy, DoR, patient fitness, access, toxicity and reimbursement.
- Belantamab combinations were viewed positively with advisors sharing growing clinical experience and interest in future mezigdomide and belantamab combination.
- Bispecifics and CAR T were recognised as highly effective options, but concerns remain around infections, CNS events, ICANS, opportunistic infections and real-world safety.
- Selinexor was considered underutilised but valuable when used carefully, with a “go low and go slow” approach and attention to cytopenia management.
- Regional guidelines, NCCN guidelines, Gulf recommendations, insurance coverage and drug availability all strongly influence real-world treatment decisions.

ADCs: antibody-drug conjugates; **CAR T-cell:** chimeric antigen receptor T-cell therapy; **CELMoDs:** cereblon E3 ligase modulators; **EMD:** extramedullary disease; **RRMM:** relapsed/refractory multiple myeloma.

Session 2: CELMoDs (Overview and recent data)

CELMoDs development program in MM

- **Broad clinical development strategy:** CELMoD trials span the MM treatment continuum, from post-transplant maintenance to heavily pretreated RRMM.

- **Key trials:**
  - **EXCALIBER:** Iberdomide vs lenalidomide in post-ASCT maintenance; iberdomide-based RRMM cohort with daratumumab + dexamethasone.
  - **SUCCESSOR-1:** Mezigdomide + bortezomib/dexamethasone vs PVd in lenalidomide-exposed RRMM.
  - **SUCCESSOR-2:** Mezigdomide + carfilzomib/dexamethasone vs carfilzomib/dexamethasone in lenalidomide- and CD38-exposed RRMM.

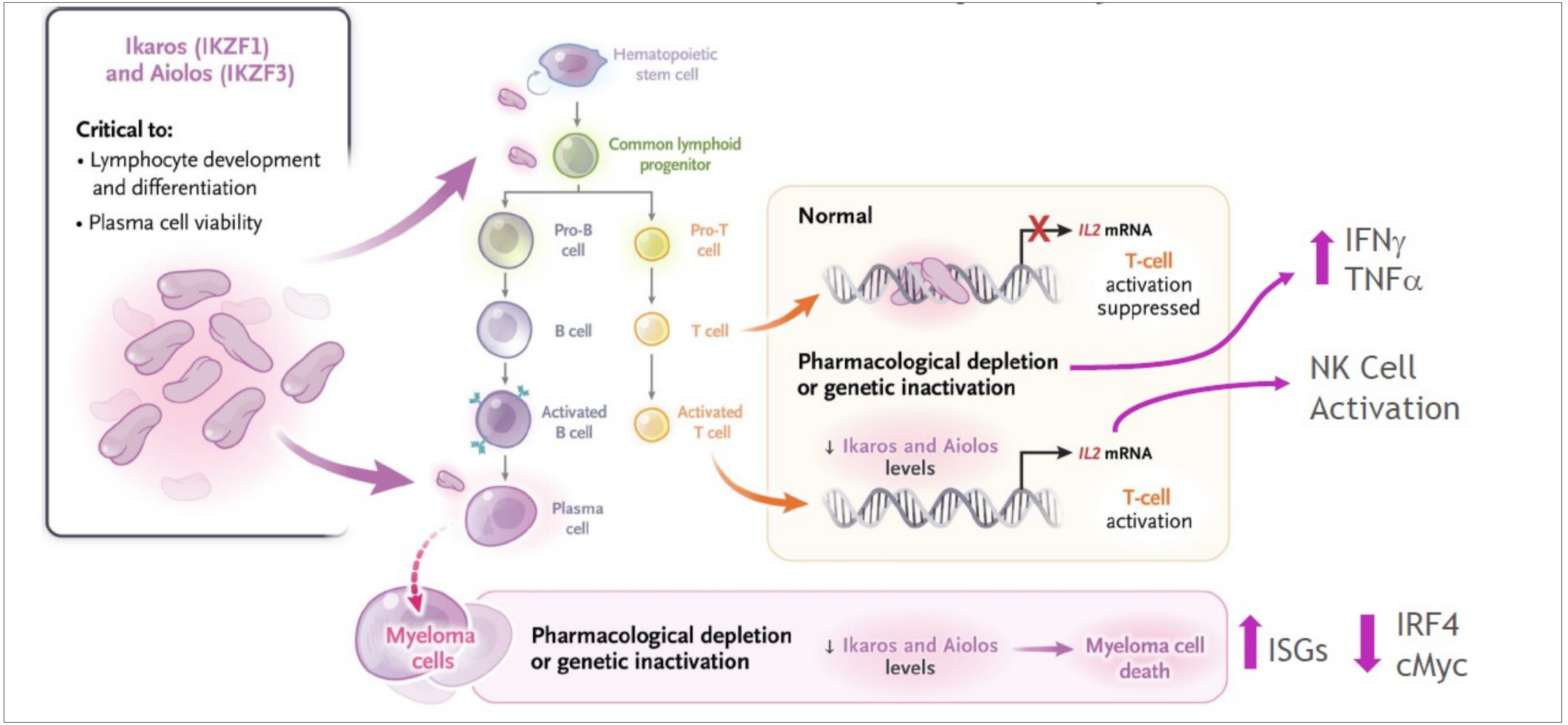
- **Expanding evidence base:** ~40 CELMoD trials are evaluating combinations across diverse patient populations, including with T-cell redirecting therapies.

ASCT: autologous stem cell transplantation; **CD38:** cluster of differentiation 38; **CELMoDs:** cereblon E3 ligase modulators; **MM:** multiple myeloma; **PVd:** pomalidomide, bortezomib, and dexamethasone; **RRMM:** relapsed/refractory multiple myeloma; **T-cell:** thymus-derived lymphocyte.

Lonial S et al. Future Oncol. 2025;21(14):1761-1769; Richardson PG et al. J Clin Oncol. 2026;44(17_suppl).

Role of Aiolos and Ikaros in MM

CELMoDs target IKZF1 (Ikaros) and IKZF3 (Aiolos) to enhance anti-myeloma activity and immune modulation

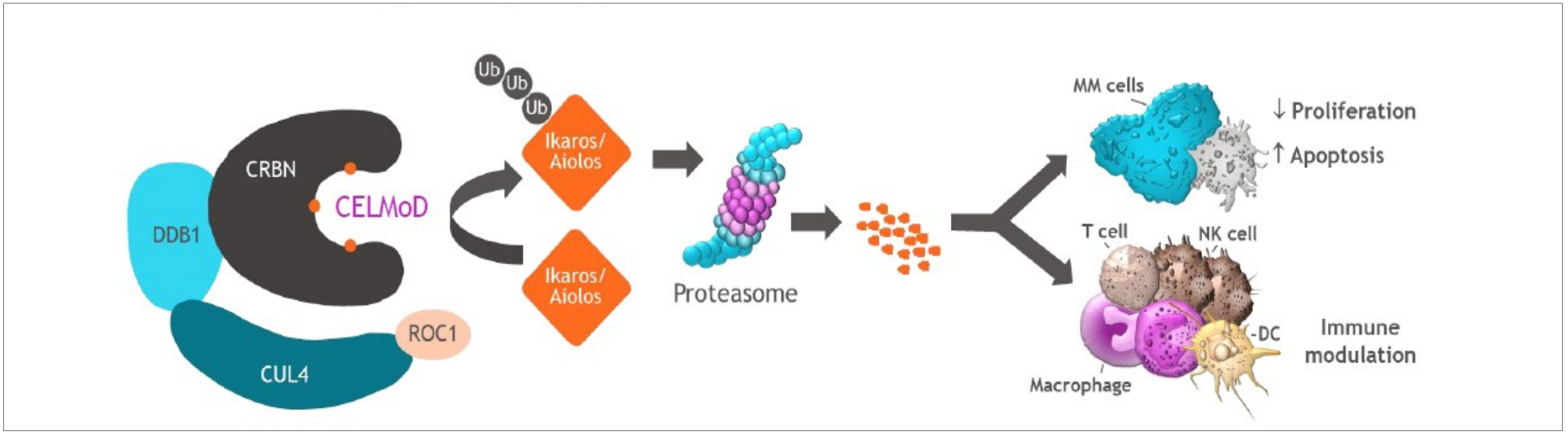


cMyc: cellular myelocytomatosis oncogene; **CELMoDs:** cereblon E3 ligase modulators; **IFNγ:** interferon gamma; **IKZF1:** ikaros family zinc finger protein 1 (ikaros); **IKZF3:** ikaros family zinc finger protein 3 (aiolos); **IL-2:** interleukin-2; **IRF4:** interferon regulatory factor 4; **ISGs:** interferon-stimulated genes; **MM:** multiple myeloma; **mRNA:** messenger ribonucleic acid; **NK:** natural killer; **TNFα:** tumor necrosis factor alpha.

CELMoD agents in MM

Iberdomide and Mezigdomide are oral, potent CELMoD compounds that induce rapid and deep degradation of Ikaros and Aiolos, hemopoietic-specific transcription factors important for regulating lymphocyte differentiation

- Loss of Aiolos and Ikaros is associated with tumoricidal and immune stimulatory effects in MM models.
- In pre-clinical models, CELMoDs compared with IMiDs: **(1)** bind cereblon with unique interactions and higher affinity, **(2)** induce degradation of Aiolos and Ikaros more potently and efficiently **(3)** have greater tumoricidal and immune stimulatory activity and **(4)** overcome LEN- and POM-resistance.



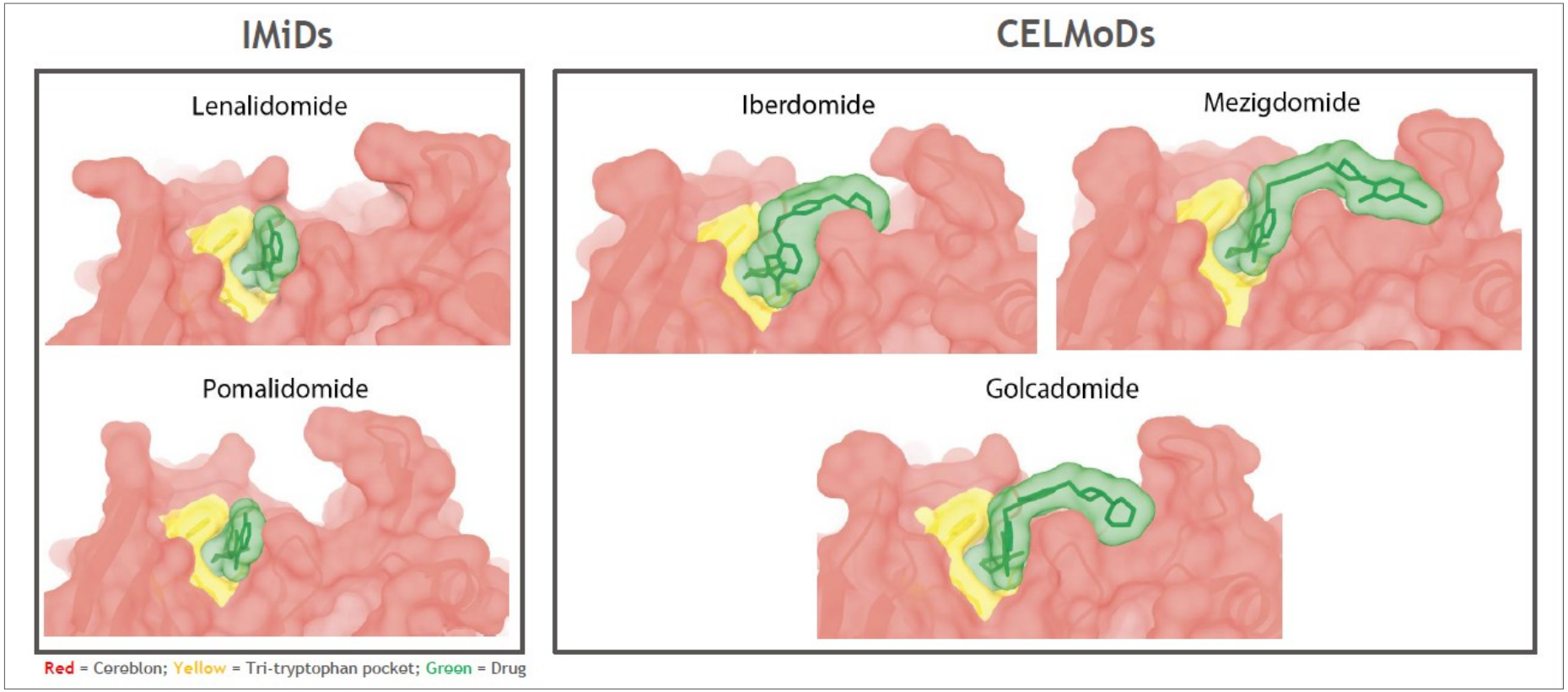
CELMoD: cereblon E3 ligase modulator; **CRBN:** cereblon; **CUL4:** cullin 4; **DC:** dendritic cell; **DDB1:** DNA damage-binding protein 1; **E3:** ubiquitin protein ligase; **IBER:** iberdomide; **LEN:** lenalidomide; **MM:** multiple myeloma; **NK:** natural killer; **POM:** pomalidomide; **ROC1:** regulator of cullins-1; **Ub:** ubiquitin.

van de Donk NWCJ et al. Onco Targets Ther. 2025;18:921-933.

CELMoDs: Enhanced mechanism over IMiDs

- Lenalidomide and pomalidomide achieve only partial complex closure (~20–25%)
- Iberdomide achieves greater engagement (~50%)
- Mezigdomide achieves near-complete closure approaching (~100%)

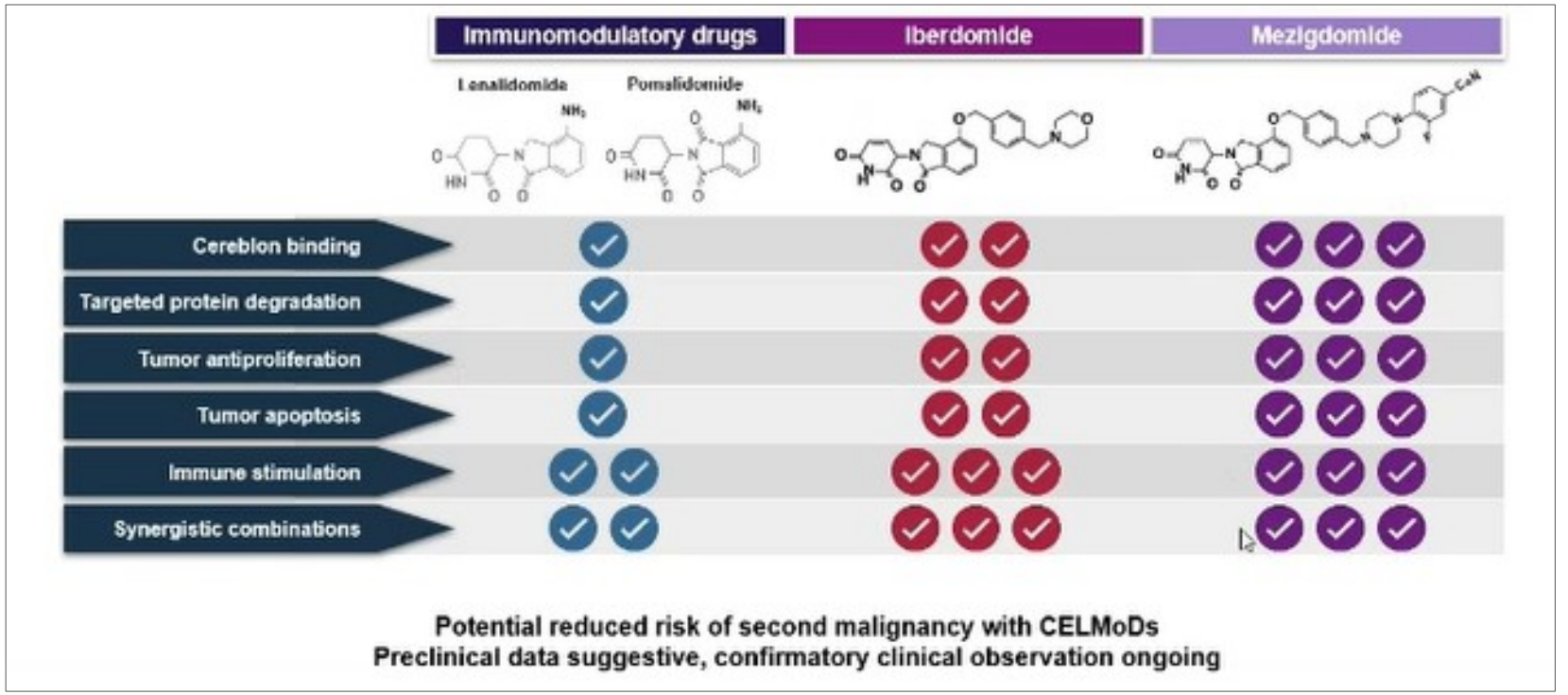
CELMoDs bind CRBN uniquely outside of tri-tryp pocket



CELMoDs: cereblon E3 ligase modulator; **CRBN:** cereblon; **IMiDs:** Immunomodulatory drugs.

Molecular advantages of CELMoDs over IMiDs

- CELMoDs (iberdomide and mezigdomide) exhibit superior CRBN binding and activation through single S-isomer design, enabling enhanced Ikaros/Aiolos degradation compared with IMiDs.
- Greater CRBN conformational activation drives faster and deeper protein degradation, translating into stronger anti-myeloma activity and efficacy in IMiD-resistant models.
- Mezigdomide demonstrated the broadest preclinical activity profile, including tumour cell killing, immune stimulation, and combination potential, supporting its clinical development in RRMM.

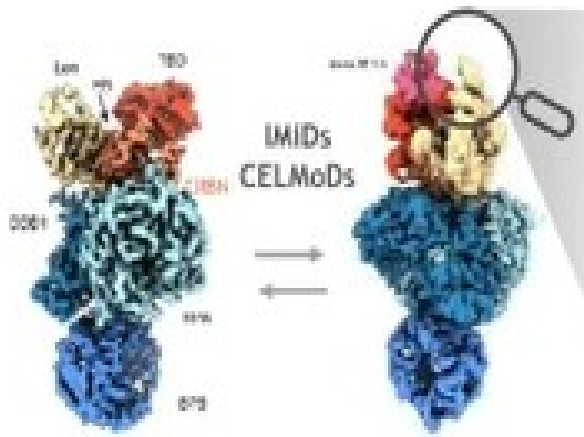


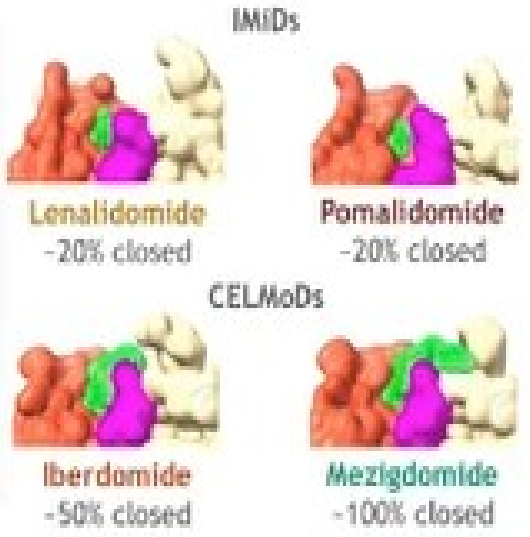
CRBN: cereblon; **CELMoDs**: Cereblon E3 ligase modulators; **IMiDs:** immunomodulatory drugs; **RRMM: r**elapsed/refractory multiple myeloma.
Hartley-Brown MA et al. Cancers (Basel). 2024;16(6):1166.

CELMoDs are a distinct class of protein Degraders, not simply next-generation IMiDs

	Compound
Classic IMiD Agents These agents exist as a mixture of S and R isomers, of which the R isomer is known to bind less efficiently to cereblon[1]	Thalidomide[2]
	Lenalidomide[2]
	Pomalidomide[2]
Novel CELMoD™ Agents These agents are administered as a single S isomer and bind to cereblon with greater potency[1,2]	Iberdomide[2]
	Mezigdomide[1]

Novel CELMoD™ agents are biochemically distinct molecules that bind to cereblon with higher specificity and affinity than classic IMiD agents[3,4]



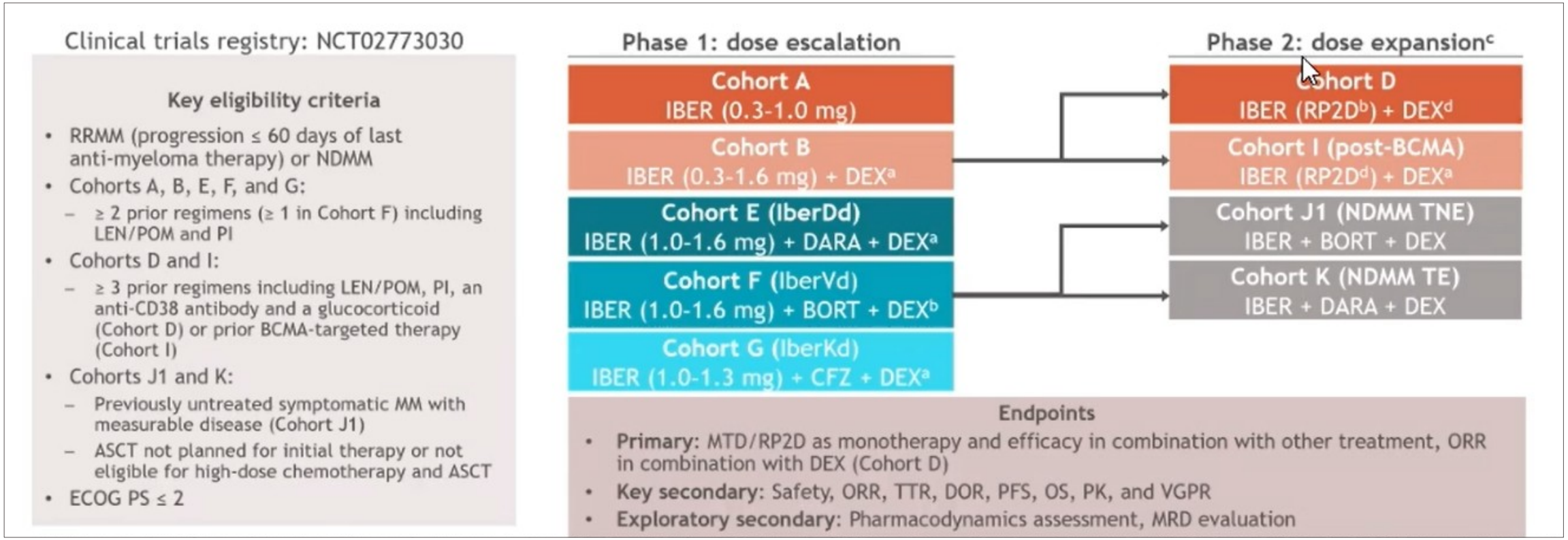


	LEN[4]	POM[3]	IBER[3,4]	MEZI[3]
Percent Closed	25%	20%	50%	100%

Preclinical data have shown that the ability to bind to cereblon and induce a **closed, active conformation** leads to substrate degradation and downstream direct anti-tumor effects[4]

Aiolos: ikaros family zinc finger protein 3 (IKZF3); **CRBN:** cereblon; **CELMoDs:** cereblon E3 ligase modulators; **IMiDs:** immunomodulatory drugs; **IBER**: iberdomide; **LEN:** lenalidomide; **POM:** pomalidomide; ; **POM:** pomalidomide; **MEZI:** mezigdomide.

CC-220-MM-001 (Phase 1b/2a): Study design



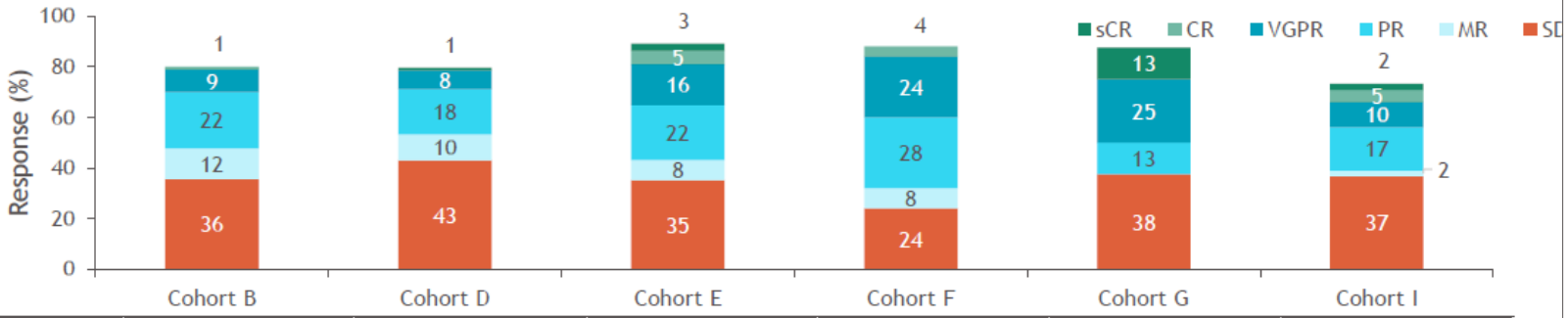
aDEX given at a dose of 40 mg (20 mg in patients aged > 75 years) on days 1, 8, 15, and 22 of each 28-day cycle; bIBER was given at 1.0, 1.3, or 1.6 mg on D1-14 in C1-8, and D1-21 in C ≥ 9, BORT was started at starting at 1.3 mg/m2 on D1, 4, 8, and 11 in C1-8, and DEX was given at a dose of 20 mg on D1, 2, 4, 5, 8, 9, 11, and 12 in C1-8 (10 mg in patients > 75 years of age), and 40 mg weekly in C ≥ 9 (20 mg in patients > 75 years); cCohort C (IBER monotherapy expansion) was planned, but not opened; d1.6 mg once a day.

ASCT: autologous stem cell transplant; **BCMA:** B-cell maturation agent; **BORT:** bortezomib; C, cycle; **CFZ:** carfilzomib; D, day; **DARA:** daratumumab; **DEX:** dexamethasone; **DOR:** duration of response; **ECOG PS:** Eastern Cooperative Oncology Group performance status; **IBER:** iberdomide; **IberDd:** iberdomide, daratumumab, and dexamethasone; **IberKd:** iberdomide, carfilzomib, and dexamethasone; **IberVd:** iberdomide, bortezomib, and dexamethasone; **LEN:** lenalidomide; **MM:** multiple myeloma; **MRD:** minimal residual disease; **MTD:** maximum tolerated dose; **NDMM:** newly diagnosed multiple myeloma; **ORR:** overall response rate; **OS:** overall survival; **PFS:** progression-free survival; **PI:** proteasome inhibitor; **PK:** pharmacokinetics; **POM:** pomalidomide; **PS:** performance status; **RP2D:** recommended phase 2 dose; **RRMM:** relapsed/refractory multiple myeloma; **TE:** transplant eligible; **TNE:** transplant non-eligible; **TTR:** time to response; **VGPR:** very good partial response.

ClinicalTrials.gov. https://ClinicalTrials.gov/ct2/show/NCT02773030. Accessed February 11, 2026; CC-220-MM-001 Study Protocol. Celgene Corporation. 2020.

Clinical outcomes with iberdomide in RRMM

- Iberdomide demonstrated activity in a **highly refractory population** (median 4-7 prior lines; high triple-class and IMiD-refractory rates), with **ORR 26-56% and disease control rates up to 88-89%.**
- **Iberdomide + daratumumab (IberDd)** showed promising efficacy in late-line RRMM, with **ORR 45.9%, clinical benefit rate 54.1%, and disease control rate 89.2%.**
- Responses were durable, with **63% ongoing at 12 months and 37% at 24 months**; mezigdomide retained activity against CRBN A365D mutant clones associated with pomalidomide resistance.



Cohort	Cohort B[1] (n = 90)	Cohort D[1,2] (n = 107)[a]	Cohort E[3] (n = 37)[b]	Cohort F[3] (n = 25)[b]	Cohort G[3] (n = 8)[b]	Cohort I[4] (n = 41)
Regimen	IBER + DEX	IBER + DEX	IberDd	IberVd	IberKd	IBER + DEX[c]
Median (range) number of prior therapies, n	5 (2-12)	6 (3-23)	4 (2-13)	5 (1-14)	6 (2-8)	7 (4-15)
ORR, %	32	26	46	56	50	34
CBR, %	–	36	54	64	50	–
DCR, %	–	79	89	88	88	–

Percentages may not add up to the ORR, CBR, or DCR due to rounding. Bars for Cohorts B and D adapted with permission from Lonial S, et al. *Lancet Haematol* 2022;9:e822-e832.
[a]2 patients in SD and MR discontinued treatment because of death due to COVID-19; [b]Excludes treated patients who did not reach any post-baseline efficacy assessment and still on treatment at the time of cutoff; [c]Post BCMA; [d]Efficacy-evaluable patients (n = 16).
BCMA, B-cell maturation agent; CBR, clinical benefit rate; CR, complete response; DCR, disease control rate; DEX, dexamethasone; IBER, iberdomide; IberDd, iberdomide, daratumumab, and dexamethasone; IberKd, iberdomide, carfilzomib, and dexamethasone; IberVd, iberdomide, bortezomib, and dexamethasone; MM, multiple myeloma; MR, minimal response; ORR, overall response rate; PR, partial response; sCR, stringent complete response; SD, stable disease; VGPR, very good partial response.
1. Lonial S, et al. *Lancet Haematol* 2022;9:e822-e832; 2. Lonial S, et al. Oral presentation at ASH; December 11-14, 2021; Atlanta, GA, USA. Abstract 162; 3. Lonial S, et al. Oral presentation at EHA; June 9-17, 2021; Virtual. Abstract S187; 4. Lonial S, et al. Poster presentation at ASH; December 10-13, 2022; New Orleans, LA, USA. Abstract 1918.

A365D: alanine to aspartic acid substitution at amino acid 365 (CRBN A365D mutation); **CBR:** clinical benefit rate; **CR:** complete response; **CRBN:** cereblon; **DCR:** disease control rate; **DEX:** dexamethasone; **IBER:** iberdomide; **IberDd:** iberdomide, daratumumab, and dexamethasone; **IberKd:** iberdomide, carfilzomib, and dexamethasone; **IberVd:** iberdomide, bortezomib, and dexamethasone; **IMiD:** immunomodulatory drug; **MR:** minimal response; **ORR:** overall response rate; **PR:** partial response; **RRMM:** relapsed/refractory multiple myeloma; **sCR:** stringent complete response; **SD:** stable disease; **VGPR:** very good partial response.

IberDd Efficacy and Safety

- **IberDd (iberdomide 1.0, 1.3, or 1.6 mg + daratumumab + dexamethasone)** achieved a sustained ORR of 94.7%, with ≥CR increasing from 22.7% to 44.0%, 57.3%, and 68.0% across 7, 11, 14, and 22 months of follow-up; MRD negativity (10^{-5}) reached 64.0%, including 56.0% MRD-negative CR.

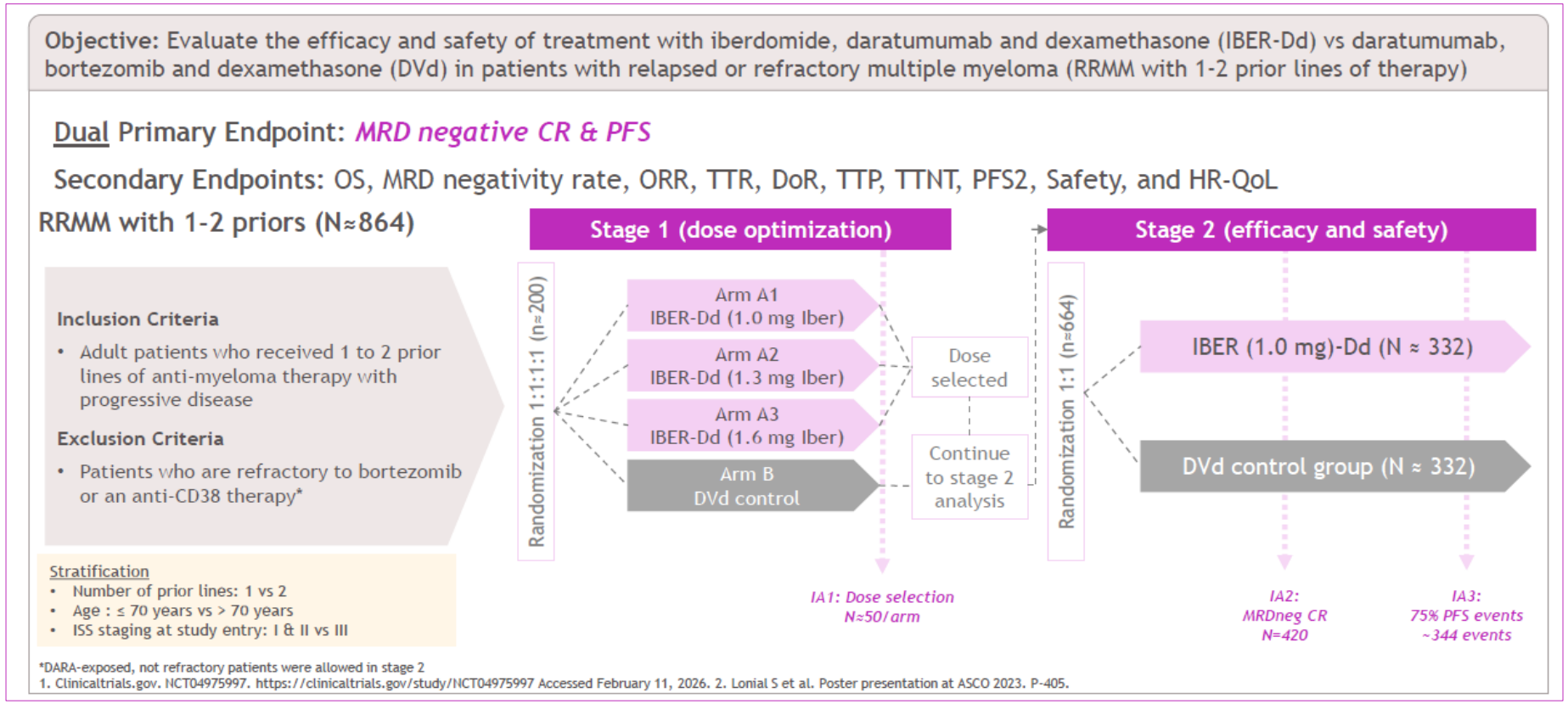
- Compared with **DRd (MAIA)** and **IsaRd (BENEFIT)**, IberDd achieved the highest ORR with ≥CR (94.7%; ≥CR 68.0%) and highest MRD negativity (64.0%), with lower rates of grade 3/4 fatigue, diarrhoea, and grade 3/4 rash (1.3% vs 7% with IsaRd).

- **Hematologic toxicities** included neutropenia (82.7%; grade 4: 44.0%), anaemia (38.7%), and thrombocytopenia (21.3%); nonhematologic TEAEs were generally low grade with no grade 3/4 diarrhoea, supporting advancement into EXCALIBER-RRMM.

CR: complete response; **DRd:** daratumumab, lenalidomide, and dexamethasone; **IsaRd:** isatuximab, lenalidomide, and dexamethasone; **MRD**: minimal residual disease; **ORR:** overall response rate; **RRMM:** relapsed/refractory multiple myeloma; **TEAEs:** treatment-emergent adverse events.

EXCALIBER-RRMM (CC-220-MM-002): IBER-Dd vs DVd

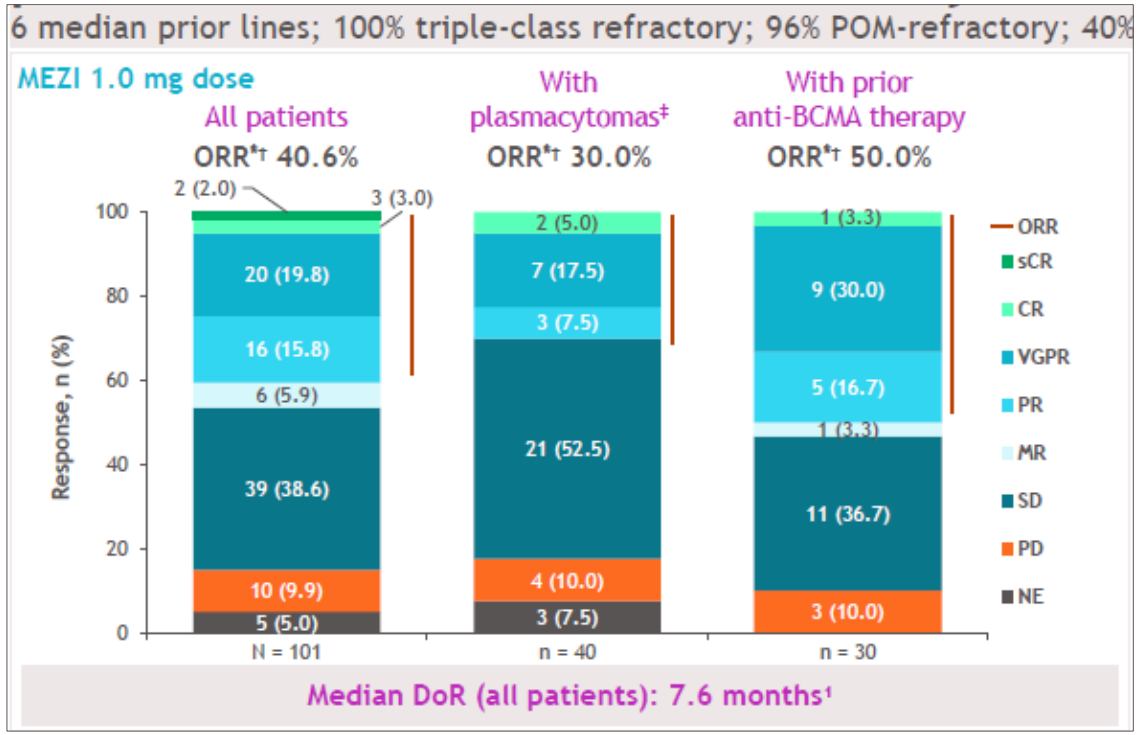
- Iberdomide has been shown to have synergy with dexamethasone and daratumumab *in vitro*.
- In a phase I/II trial, IberDd was well tolerated and demonstrated promising preliminary efficacy in patients with heavily pretreated RRMM, including patients refractory to IMiD agents, daratumumab, and PIs.
- Outcomes of EXCALIBER-RRMM may establish IberDd as a transformative treatment in early-line RRMM and potentially broaden its role in combination regimens across the disease spectrum.



CC-220-MM-002: clinical trial protocol code for the EXCALIBER-RRMM study; **CD38:** cluster of differentiation 38; **CR:** complete response; **DARA:** daratumumab; **Dd:** daratumumab and dexamethasone; **DoR:** duration of response; **DVd:** daratumumab, bortezomib, and dexamethasone; **EXCALIBER-RRMM:** exploratory clinical trial of iberdomide in relapsed/refractory multiple myeloma (trial name); **HR-QoL:** health-related quality of life; **IA1:** interim analysis 1; **IA2:** interim analysis 2; **IA3:** interim analysis 3; **IBER:** iberdomide; **IberDd:** iberdomide, daratumumab, and dexamethasone; **IMiD:** immunomodulatory drug; **ISS:** international staging system; **MRD:** minimal residual disease; **ORR:** overall response rate; **OS:** overall survival; **PFS:** progression-free survival; **PFS2:** second progression-free survival; **PI:** proteasome inhibitor; **RRMM:** relapsed/refractory multiple myeloma; **TTP:** time to progression; **TTNT:** time to next treatment; **TTR:** time to response.
Lonial S et al. Future Oncol. 2025;21(14):1761-1769.

CC-92480-MM-001 dose-expansion cohort: MEZI + DEX oral doublet demonstrated efficacy in RRMM patients with multi-class refractory disease

- **CC-92480-MM-001:** Mezigdomide 1 mg (Days 1–21) + dexamethasone evaluated in heavily pretreated triple-class refractory RRMM (median 6 prior lines).

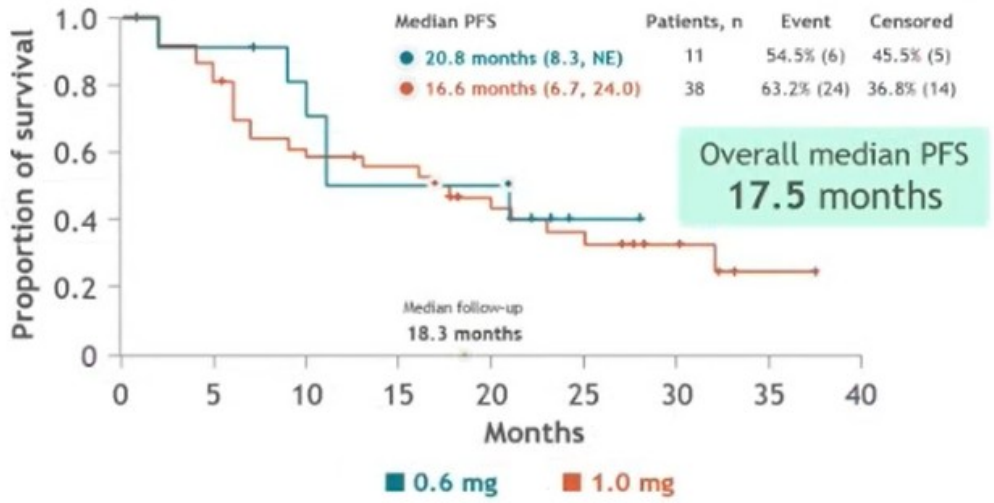


Mezigdomide is not currently approved in any market and has not been evaluated by any regulatory agency as safe or effective for multiple myeloma.
*Data cutoff: 16 September 2022. †PR or better‡. ‡ncluding extramedullary soft tissue only disease as well as soft tissue bone-related plasmacytomas.

BCMA: B cell maturation antigen; **CR:** complete response; **DEX:** dexamethasone; **DoR:** duration of response; **EMD:** extramedullary disease; **IMiD:** immunomodulatory drug; **MEZI:** mezigdomide; **MR:** minimal response; **NE:** not evaluable; **ORR:** overall response rate; **PD:** progressive disease; **PET:** positron emission tomography ; **PR:** partial response; **RRMM:** relapsed/refractory multiple myeloma; **sCR:** stringent complete response; **SD:** stable disease; **VGPR:** very good partial response.
1. Richardson PG et al. Oral presentation at ASH 2022. Abstract 568; 2. Richardson PG et al. Oral presentation at ASCO 2020. Abstract 8500; 3. Data on file. Princeton, NJ: Bristol-Myers Squibb Company.

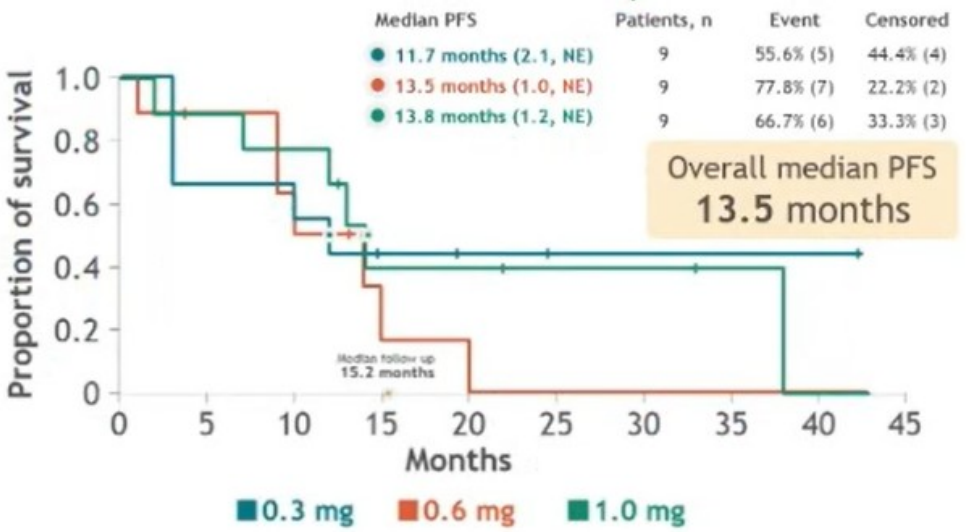
MeziVd and MeziKd efficacy summary

PFS - Cohort D (MeziVd)
(prior anti-CD38 exp/ref = 38.8%/34.7%)



	All doses	1.0 mg
ORR,[a] % (95% CI)	85.7 (72.8-94.1)	84.2 (68.7-94.0)
DOR, median (95% CI), months	19.4 (9.7-NA)	19.4 (7.0-NA)

PFS - Cohort C (MeziKd)
(prior anti-CD38 exp/ref = 77.8%/74.1%)



	All doses	1.0 mg
ORR,[b] % (95% CI)	85.2 (66.3-95.8)	77.8 (40.0-97.2)
DOR, median (95% CI), months	11.9 (6.4-35.9)	11.9 (0.2-NA)

Data cutoff: June 28, 2024.aORR refractory to LEN and anti-CD38 mAbs (data cutoff: May 9, 2024): cohort D = 75.0%. bMedian number of prior therapies = 2. eORR refractory to LEN and anti-CD38 mAbs (data cutoff: May 9, 2024): cohort C = 82.4%.

CD: clusters of differentiation; **DOR:** duration of response; **EHA:** European Hematology Association; **LEN:** lenalidomide; **mAb:** monoclonal antibody; **Mezi:** mezigdomide; **NE:** not evaluable; **ORR:** overall response rate; **PFS:** progression-free survival; **MeziKd:** mezigdomidea, carfilzomib, and dexamethasone; **MeziVd:** mezigdomide, bortezomib, and dexamethasone; **RRMM:** relapsed/refractory multiple myeloma.

1.Sandhu I et al. Presented at EHA Annual Meeting 2025. Abstract PF723.

SUCCESSOR-1 and SUCCESSOR-2 trial overview

SUCCESSOR-1* compares MeziVd vs PVd in early RRMM (1–3 prior lines), enrolling lenalidomide-exposed, pomalidomide-naïve, and bortezomib-non-refractory patients; PFS is the primary endpoint (OS key secondary endpoint).

SUCCESSOR-2 (CA057-008) evaluates MeziKd vs Kd in RRMM after ≥1 prior line, enrolling ~606 patients with prior lenalidomide and anti-CD38 exposure (excluding prior mezigdomide/carfilzomib); PFS is the primary endpoint and OS the key secondary endpoint.

→

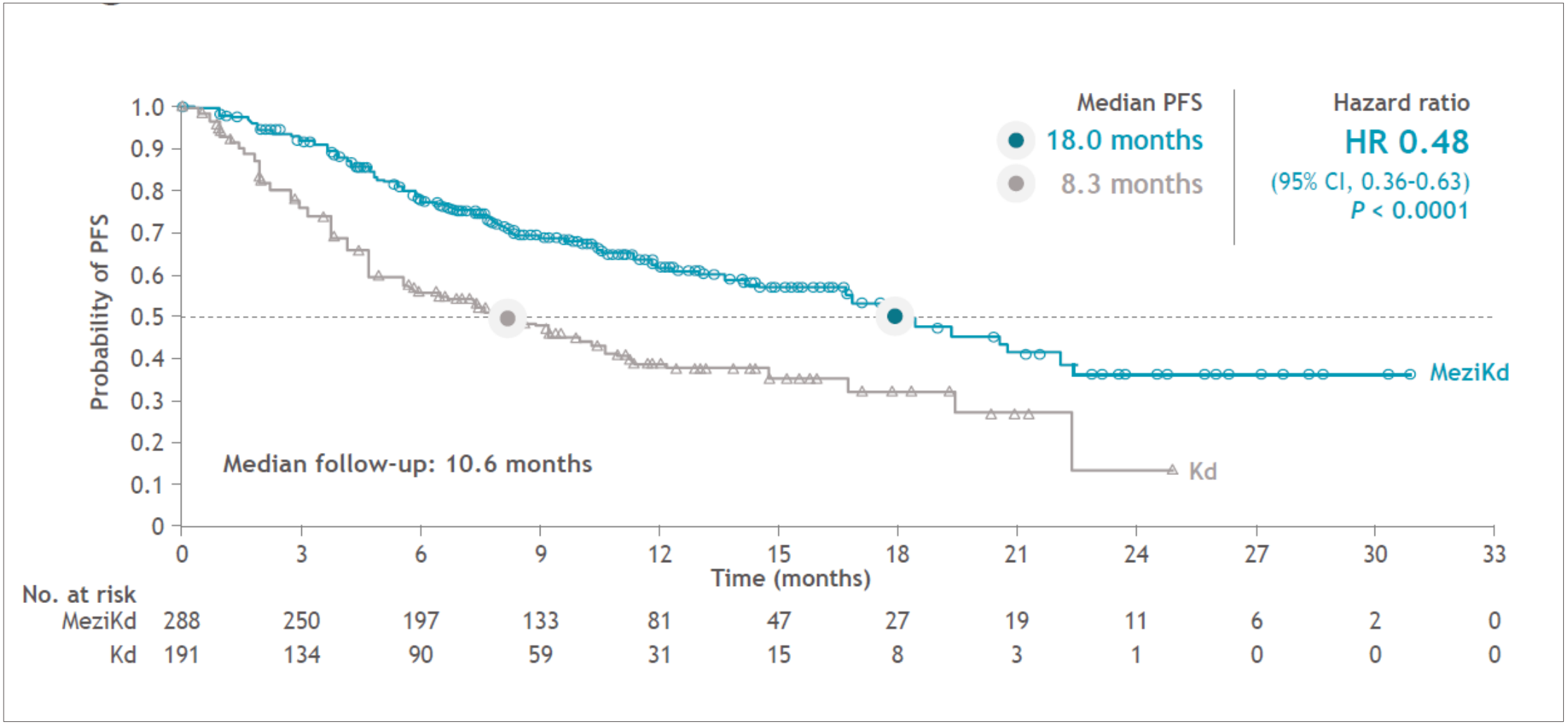
In **SUCCESSOR-2**, patients had a median of 2 prior lines across 26 countries; 52.4% remained on MeziKd versus 31.4% on Kd, with lower discontinuation due to disease progression (29.5% vs 48.7%) and comparable median follow-up (10.5–10.8 months).

*Results are expected to read out in **2027**

Anti-CD38: anti-CD38 monoclonal antibody; **Kd:** carfilzomib and dexamethasone; **MeziKd:** mezigdomide, carfilzomib, and dexamethasone; **MeziVd:** mezigdomide, bortezomib, and dexamethasone; **OS:** overall survival; **PFS:** progression-free survival; **PVd:** pomalidomide, bortezomib, and dexamethasone; **RRMM:** relapsed/refractory multiple myeloma.

Richardson PG et al. J Clin Oncol. 2026;44(17_suppl).

SUCCESSOR-2: Progression-free survival (MeziKd vs Kd)



CI: confidence interval; **HR:** hazard ratio; **Kd:** carfilzomib and dexamethasone; **MeziKd:** mezigdomide, carfilzomib and dexamethasone; **PFS:** progression-free survival.

Richardson PG et al. J Clin Oncol. 2026;44(17_suppl).

SUCCESSOR-2: MeziKd efficacy and safety

MeziKd more than doubled median PFS versus Kd, with benefit observed across key subgroups, including patients ≥75 years (HR 0.52), lenalidomide-/anti-CD38-refractory disease (HR 0.47), and extraosseous EMD (HR 0.33)

PFS2 was 23.6 vs 13.0 months (HR 0.53); fewer patients received subsequent antimyeloma therapy (24.0% vs 49.7%), including BsAbs (10.4% vs 44.2%) and pomalidomide (5.6% vs 36.8%)

ORR was 80.2% with MeziKd vs 53.4% with Kd; OS showed HR 0.79 at a median follow-up of 10.6 months

Grade 3/4 AEs were more frequent with MeziKd (83.7% vs 56.5%), mainly neutropenia (61.1% vs 9.1%), thrombocytopenia (39.2% vs 22.6%), and anemia (26.0% vs 15.1%)

AEs: Adverse events; **Anti-CD38:** anti-CD38 monoclonal antibody; **BsAbs:** bispecific antibodies; **CI:** confidence interval; **EMD:** extramedullary disease; **HR:** hazard ratio; **Kd:** Carfilzomib and dexamethasone; **MeziKd:** Mezigdomide, carfilzomib, and dexamethasone; **ORR:** overall response rate; **OS:** overall survival; **PFS:** progression-free survival; **PFS2:** progression-free survival after next line of therapy.

Richardson PG et al. J Clin Oncol. 2026;44(17_suppl).

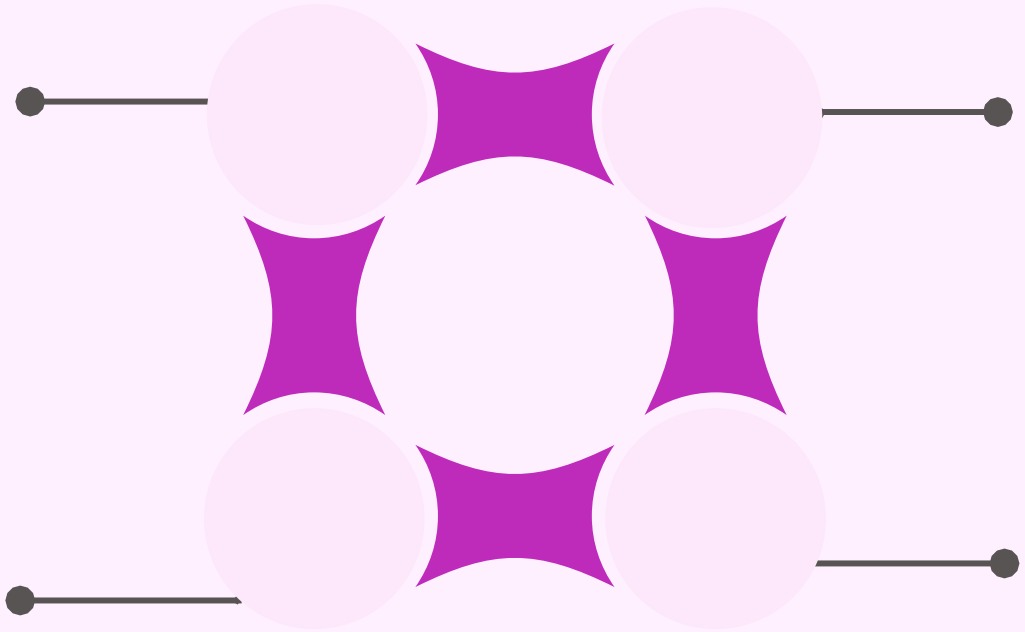
Discussion 2: Perspectives on CELMoDs

How should the efficacy and safety of IberDd be interpreted compared with currently available second- and third-line treatment options?

Views on efficacy and safety of IberDd

Demonstrate manageable hematological toxicity with favorable neurological and gastrointestinal tolerability

Neutropenia can be managed with G-CSF and dose modification



Dosing generally ranges from 0.75 to 1.3 mg, depending on the regimen and tolerability

Follow-up and comparative data are needed to define its positioning versus belantamab-based therapies

G-CSF: granulocyte colony-stimulating factor; **IberDd:** iberdomide, daratumumab, and dexamethasone.

How do CELMoDs differ from traditional IMiDs?

CELMoDs vs IMiDs

- ✓ CELMoDs are a distinct therapeutic class, not simply next-generation IMiDs, and may remain effective in lenalidomide- or pomalidomide-refractory patients.
- ✓ Oral iberdomide combined with monthly subcutaneous daratumumab could offer a convenient, fixed treatment schedule.
- ✓ Anti-CD38 maintenance remains important, although iberdomide-based combinations may influence its future duration and role.
- ✓ Iberdomide may offer improved tolerability, lower secondary malignancy risk, and greater efficacy, including in high-risk disease.

Anti-CD38: anti-cluster of differentiation 38; **CELMoDs:** cereblon E3 ligase modulators; **CD38:** cluster of differentiation 38; **IMiDs:** immunomodulatory drugs.

How should the efficacy, safety and clinical positioning of mezigdomide be interpreted?

Views of efficacy and safety of mezigdomide

- ✓ Mezigdomide demonstrated encouraging efficacy and a manageable safety profile, although longer follow-up and broader clinical experience are needed.
- ✓ Neutropenia and infection risk appear manageable with proactive supportive care, including G-CSF, PCP prophylaxis, close monitoring and early intervention.
- ✓ Regional differences in infection rates may reflect variations in BMI, pharmacokinetics, supportive care and healthcare practices, requiring further investigation.
- ✓ CELMoDs may provide a practical option where TCEs or CAR T-cell therapy are inaccessible, with mezigdomide–ADC combinations viewed as a promising future strategy.

ADC: antibody-drug conjugate; **BMI:** body mass index; **CAR T-cell:** chimeric antigen receptor T-cell; **CELMoDs:** cereblon E3 ligase modulators; **G-CSF:** granulocyte colony-stimulating factor; **PCP:** *Pneumocystis jirovecii* pneumonia; **TCEs:** T-cell engagers.

Which patient profiles are most suitable for IberDd or mezigdomide-based combinations, and how should CELMoDs be sequenced relative to ADCs and T-cell-engaging therapies?

Patient selection and sequencing of CELMoD-based therapy

- ✓ IberDd may be suitable after frontline triplet therapy, while mezigdomide-based regimens may be preferred following quadruplet exposure.
- ✓ CELMoDs could be positioned in the second- or third-line setting, particularly before bispecific antibodies or CAR T-cell therapy because of their more manageable safety profile and practical delivery.
- ✓ ADC–CELMoD combinations were viewed as a promising early-relapse strategy, especially in daratumumab- or lenalidomide-refractory patients.
- ✓ CELMoDs may be particularly valuable where access to CAR T-cell therapy or bispecific antibodies is limited.
- ✓ Cytopenias are generally manageable with supportive care, and IVIG requirements appear lower than with T-cell–directed therapies.

CAR T: chimeric antigen receptor T-cell therapy; **CELMoDs:** Cereblon E3 Ligase Modulators; **IberDd:** iberdomide, daratumumab, and dexamethasone; IVIG: intravenous immunoglobulin.

Could CELMoDs eventually replace current treatment standards in multiple myeloma?

CELMoDs for all seasons

- ✓ Iberdomide may move into frontline therapy, potentially replacing lenalidomide because of improved tolerability, lower secondary malignancy risk and activity in high-risk disease.
- ✓ Mezigdomide is expected to be positioned strategically in relapse and may move earlier as further combination data emerge.
- ✓ ADCs may remain preferred initially because of greater clinical experience, while CELMoDs could be sequenced before TCEs, BsAbs, and CAR T-cell therapy.
- ✓ Infection risk appears manageable with proactive monitoring and PCP prophylaxis, with further evidence awaited from ongoing mezigdomide combination studies.

ADCs: antibody–drug conjugates; **BsAbs:** bispecific antibodies; **CAR T-cell therapy:** chimeric antigen receptor T-cell therapy; **CELMoDs:** cereblon E3 ligase modulators; **PCP:** *Pneumocystis jirovecii* pneumonia; **TCEs:** T-cell engagers.

Key takeaways

- Iberdomide was viewed as a distinct CELMoD rather than simply a next-generation IMiD, with potential use in lenalidomide or pomalidomide refractory patients.
- The safety profile of iberdomide was considered favourable with limited neurological toxicity and good GI tolerability.
- Neutropenia with iberdomide was considered manageable with dose adjustment and G-CSF support, while stem cell or T-cell collection is not expected to be affected.
- IberDd was considered a convenient option with oral iberdomide and monthly SC daratumumab, offering a more patient-friendly schedule.
- Mezigdomide data were considered encouraging, although longer follow-up and broader experience are needed to better define safety and infection risk.
- Infection risk with mezigdomide was viewed as manageable with proactive monitoring, G-CSF, PCP prophylaxis and early intervention.

CELMoD: cereblon E3 ligase modulator; **G-CSF:** granulocyte colony-stimulating factor; **GI:** gastrointestinal; **IberDd:** iberdomide, daratumumab, and dexamethasone; **IMiD:** immunomodulatory drug; **PCP:** *Pneumocystis jirovecii* pneumonia.

Key takeaways (Contd.)

- Advisors felt CELMoDs could be positioned before BsAbs and CAR T, given concerns around infections, toxicity, access and practicality with T-cell-directed therapies.
- ADC–CELMoD combinations, particularly belantamab plus mezigdomide, were viewed as highly promising future strategies, especially in early relapse.
- Overall, iberdomide was seen as likely to move earlier, potentially into frontline therapy, while mezigdomide may play a major role in relapse.

ADCs: antibody–drug conjugates; **BsAbs:** bispecific antibodies; **CAR T:** chimeric antigen receptor T-cell therapy; **CELMoDs:** cereblon E3 ligase modulators.

Thank You